Toward a Formal Account of Reversal Triggers: From Allostatic Regulation to Compositional-Distributional Semantics in Reversal Theory

Models of Consciousness 7, University of Copenhagen

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Intro: Reversal Theory

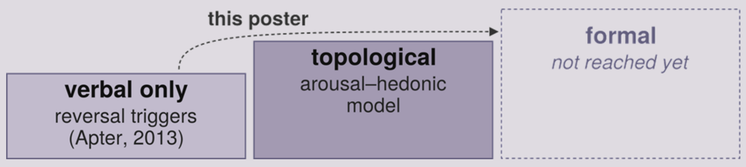
Reversal Theory (Smith & Apter, 1975) holds that motivation is **dynamic** rather than a fixed trait: people "reverse" between opposite **meta-motivational** states, depending on how they interpret a situation. These states shape how arousal is experienced: the same high arousal can feel like **anxiety** or **excitement**, depending on the state.

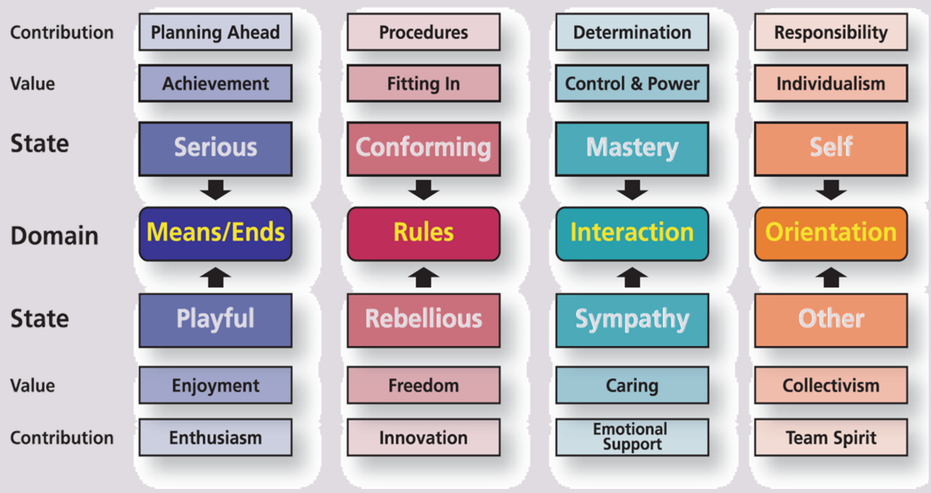
Prior Work and the Gap

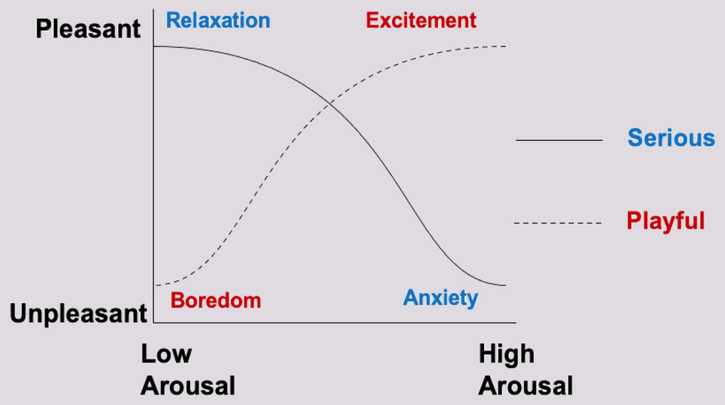
Feasibility. Using RT to weight effort in web-search records (queries, clicks, time spent) improved detection of frustrated searchers: five existing classifiers gained ~5% accuracy and ~9% precision on average (Luo et al., 2024).

Limits. Luo et al.'s method departed from RT (table; Yang, 2026).

Gap. Luo et al. did not model reversals, and RT's triggers remain at the lowest step:







Wikipedia. Reversal Theory. https://en.wikipedia.org/wiki/Reversal_Theory

Left: Four Pairs of Metamotivational States; Right: Apter's State Reversal Mechanism (the First Pair, a.k.a. Telic/Paratelic)

RT in Luo et al. (2024)	*first-person constructs, read off behavior (search logs)*
arousal	search amount (queries, clicks, time)
telic / paratelic	session topic inferred from click URLs
conformist / negativistic	how exploratory the search was

RT in Luo et al. (2024)	*RT's relations, read as self-regulated distributions or metric quantities*
arousal–hedonic curve	session count distribution over effort levels *may reflect self-regulation toward the curve's "sweet spot", but is not the curve itself*
topological relations	positions compared across diagrams

Base model: EPIC on a learning network

EPIC (Barrett & Simmons, 2015) holds that interoceptive experience is mostly top-down prediction constrained by bodily signals. We map it onto Predictive Forward-Forward learning (PFF; Ororbia & Mali, 2023), adapted toward human allostasis: error reduced by **revising predictions** or **acting on the body** (Adams et al., 2013).

Open: how RT's bistable state pairs attach to the base model is not yet settled.

Candidates: 1. two stacked networks, each a full base model: emotional category → state pairs → x; 2. adding a readout from one middle layer; 3. inferred from or read out together with core affect; 4. two separate networks for a pair of states, only one active. **Question:** how to avoid essentialism, i.e. avoid treating each state as a fixed natural kind? Happy to discuss.

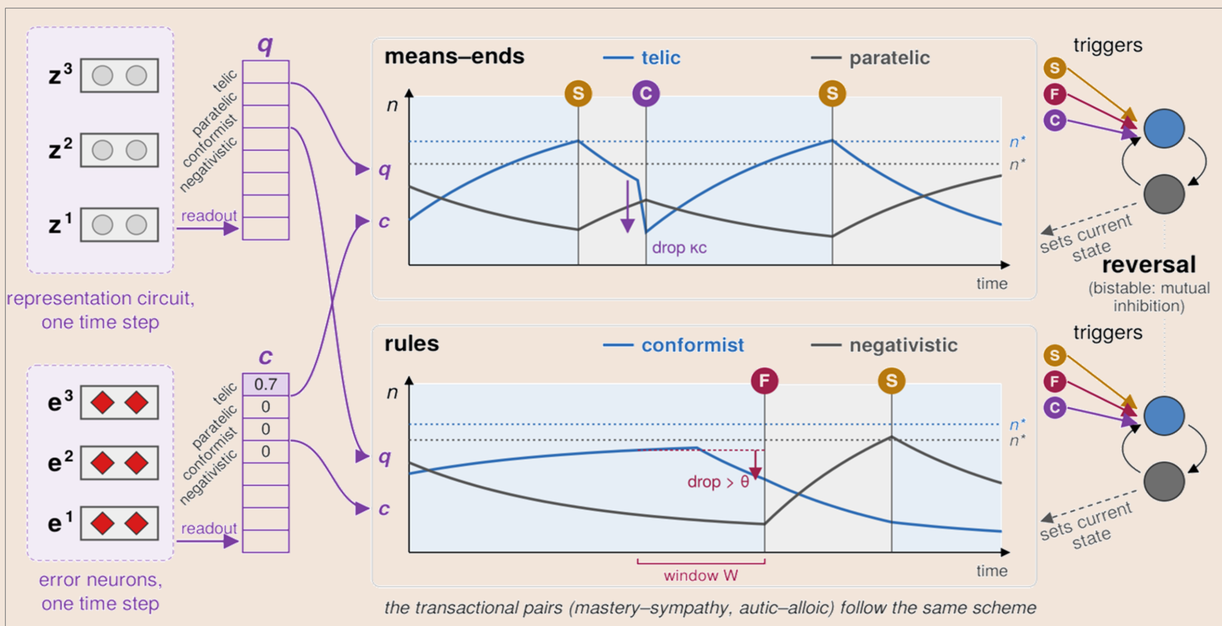
Reversal Triggers in Body Budgeting

Each of the eight states m has a need, whose satisfaction level is n_m, and target level is n^*_m.

- **Contingent events:** an outside event c_m calls for a particular state m, e.g. a sudden threat calls for the telic state. For each m, c_m gives an event strength between 0 (no event) and 1.
- **Update:** n_m decays at rate δ ($0 < \delta < 1$; for simplicity, the same δ for all states). For each time step, it is refilled by supply q_m and lowered by κc_m, where $\kappa > 0$ is the drop caused by an event of strength 1.
- **Triggers** for leaving the current state m toward its partner $\bar{m}$:

  S, satiation: n_m rises above its target n^*_m; **F, frustration:** n_m falls by more than θ_m within a recent window; **C, contingent event:** $c_{\bar{m}} > 0$.

For person i, the parameters (δ_i, κ_i, $\theta_{m,i}$, $n^*_{m,i}$) can be drawn from a shared Gaussian population distribution.



the transactional pairs (mastery–sympathy, autic–alloic) follow the same scheme

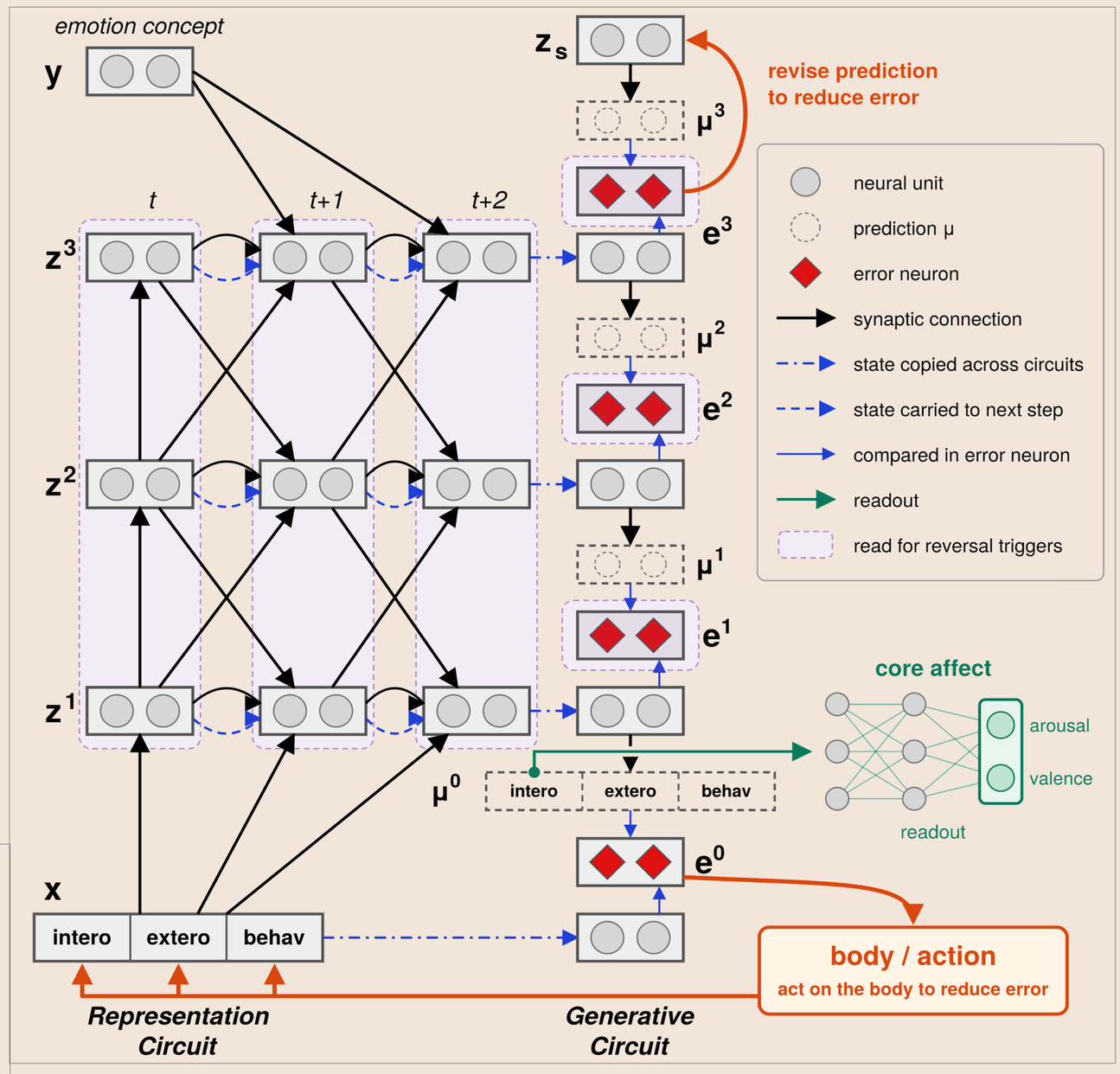
Update of the need for every state m, each step: $n_m(t) = (1 - \delta)\, n_m(t-1) + q_m(t) - \kappa\, c_m(t)$

(S) Satiation of the current state m: $n_m(t) > n^*_m$

(F) Frustration of the current state m: $n_m(t - w) - n_m(t) > \theta_m, \quad w = min(W, t - t_0)$

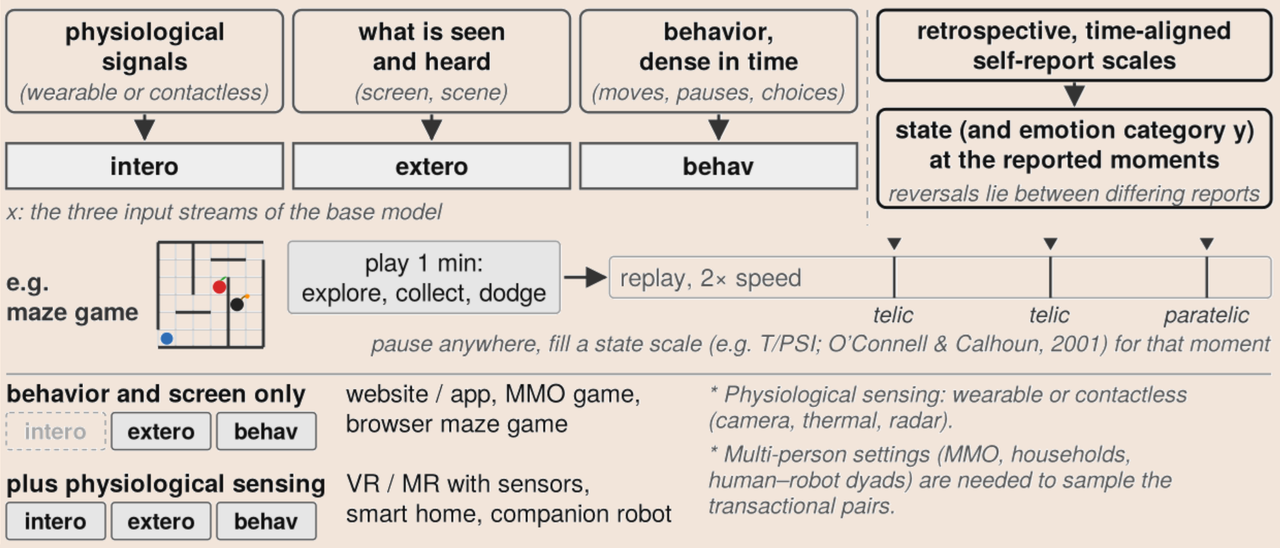
(C) Contingent event toward the partner state $\bar{m}$: $c_{\bar{m}}(t) > 0$

Reversal m → $\bar{m}$ (its partner): (S or F or C)



Toward Fitting the Triggers

The base network learns from x and y by local (PFF) rules. The weights that compute q and c, and the parameters δ, κ, θ, n*, are then fitted to each user's time-aligned self-reported states.



* *Physiological sensing: wearable or contactless (camera, thermal, radar).*

* *Multi-person settings (MMO, households, human–robot dyads) are needed to sample the transactional pairs.*

Compositional-Distributional Semantics (CDS): A First Check

Step by step (online)

n, q, c, n, θ: one entry per state (8); δ, κ: shared scalars*

$$n(t) = A\, n(t-1) + q(t) - \kappa\, c(t), \quad A = (1 - \delta)\, I$$

$$S = [\, n > n^* \,], \quad F = [\, n(t - w) - n > \theta \,], \quad C = [\, P\, c > 0 \,]$$

$(P\, v)_m = v_{\bar{m}}$: each entry takes its partner's value
I: 8 × 8 identity matrix; [·] element-wise

All steps at once (one matrix product fits a session)

$$n(t) = A^t\, n(0) + \sum_{k=0}^{t-1} A^k\, [\, q(t-k) - \kappa\, c(t-k) \,]$$

$$\begin{bmatrix} n(1) \\ \vdots \\ n(T) \end{bmatrix} = \begin{bmatrix} (1-\delta) & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ (1-\delta)^T & (1-\delta)^{T-1} & (1-\delta)^{T-2} & \cdots & 1 \end{bmatrix} \begin{bmatrix} n(0) \\ q(1) - \kappa c(1) \\ \vdots \\ q(T) - \kappa c(T) \end{bmatrix}$$

Matrix–Vector CDS and Beyond

Both forms are a leaky integrator (older input fades) and use CDS only at its simplest level: matrices acting on vectors, like adjectives acting on nouns.

- Abstract's question: can trigger inputs share one general, structured space, rather than RT's three-way split? Apter (2001, p. 46): reversal occurs when "combined forces" for it exceed those against it; the model checks S, F, C alone.
- How to combine them: in plain and weighted sums, satiation and frustration offset each other (a sudden drop lowers n, cutting satiation while raising frustration), whereas Apter's combined forces should reinforce. Fix this within a sum, or switch to products (satiation × frustration, satiation × event)?
- Measurement: in retrospective scales, earlier answers might shift later ones. Quantum cognition models such order effects (Wang & Busemeyer, 2013); since CDS uses the same maths, the two could perhaps be combined.

E-Poster, References, Author Profile: